BUSINESS ANALYTICS (BMGT107L)

VELLORE INSTITUTE OF TECHNOLOGY (VIT) UNIVERSITY

Chennai Campus | School of Computer Science and Engineering (SCOPE)

FALL SEMESTER 2026-27

DIGITAL ASSIGNMENT 3

PROJECT TITLE

Restaurant Recommendation

A Data-Driven Analytical System for Restaurant Quality Assessment, Econometric Modeling, and Decision Support

Programme:	B.Tech. Computer Science and Engineering (Data Science)	**Slot: G2 + TG2**
Course:	BMGT107L – Business Analytics	**Semester:** Fall 2026-27
Faculty In-Charge:	**Prof. BALAJI J**	**School: SCOPE**

S.No.	Student Candidate Name	Registration No.
1	**PATEL KAVYA VIPULKUMAR**	23BDS1076
2	**SAHIL KUMAR**	23BDS1099

Live Deployed Model:	https://kavyapatel83-business-analytics-restaurant-dashboard-tpt3xx.streamlit.app/
GitHub Repository:	https://github.com/KavyaPatel83/Business-Analytics

School of Computer Science and Engineering (SCOPE)

Vellore Institute of Technology, Chennai Campus, Vandalur-Kelambakkam Road, Chennai – 600127

Digital Assignment 3: Restaurant Recommendation

A Data-Driven Analytical System for Restaurant Quality Assessment, Econometric Modeling, and Decision Support

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1. Introduction and Problem Statement

Bangalore (Bengaluru), globally recognized as the technological powerhouse of India, possesses one of the most vibrant, heterogeneous, and rapidly expanding urban dining ecosystems in Asia. Characterized by high concentrations of IT professionals, corporate executives, and college students, dining out and ordering meals through digital platforms (such as Zomato and Swiggy) have become embedded into daily urban lifestyle. However, this vast commercial variety has generated acute information friction and market distortions.

Modern online food aggregators rely on naive arithmetic star ratings that suffer from severe mathematical and behavioral flaws. First, arithmetic averaging ignores sample size: a newly listed outlet with three 5-star reviews submitted by personal acquaintances displays a perfect 5.0 rating, outranking a landmark institution with a 4.3 rating backed by 3,000 verified reviews. Second, consumers erroneously equate high prices with superior culinary quality, whereas empirical data reveals that dining cost explains less than 15% of rating variance (Pearson r = 0.354). Third, restaurant entrepreneurs operate under immense capital risk, lacking econometric clarity regarding whether operational investments in table reservations, delivery logistics, or spatial clustering yield tangible customer satisfaction returns.

To resolve these interrelated challenges, this project establishes 'Restaurant Recommendation'—an end-to-end analytical framework and deployed decision-support system. The system ingests and consolidates 51,717 raw listings into 9,244 physical units, executes nine formal hypothesis tests, fits econometric regression and machine learning models, formulates an empirical Bayesian shrinkage recommender, and deploys an interactive, reactive web dashboard for live stakeholder decision-making.

Final Working Model: Live Deployment & Source Code Verification

Live Deployed Web Application: https://kavyapatel83-business-analytics-restaurant-dashboard-tpt3xx.streamlit.app/
Official GitHub Code Repository: https://github.com/KavyaPatel83/Business-Analytics
Deployment Platform: Streamlit Community Cloud (Official Platform PaaS, Global HTTPS Edge CDN, Active 24/7). Evaluators can directly interact with the live model, tune Bayesian weights, explore OLS/Logistic diagnostics, and query restaurant recommendations in real time.

2. Objectives of the Model and Research

The project rigorously fulfills six core technical, econometric, and managerial objectives:

- **Automated Data Sanitation & Deduplication Pipeline:** Clean, parse, and consolidate 51,717 raw scrape records into 9,244 unique physical restaurants, removing autocorrelation and multi-zone duplicates.
- **Inferential Hypothesis Testing Suite:** Execute 9 paired parametric and non-parametric hypothesis tests with standardized effect sizes (Cohen's d, Eta-squared, Cramer's V) to isolate genuine drivers of customer rating.
- **Econometric Quality Modeling (OLS Regression):** Estimate an Ordinary Least Squares multiple regression model to quantify the marginal ceteris paribus impact of table booking, online ordering, review volume, and spend.
- **Binary Logistic Classification (Premier Dining):** Train a logistic classifier with L-BFGS to predict high-rated venues (rating >= 4.0) with odds ratios, confusion matrix validation, and ROC-AUC evaluation.

• **Unsupervised Market Segmentation (K-Means):** Discover latent operational dining segments using standardized cost, review volume, and rating validated via Elbow and Silhouette diagnostics.

• **Production Deployment & Multi-Criteria Recommender:** Deploy an interactive Streamlit application integrating Bayesian shrinkage trusted ratings, value-for-money percentiles, and popularity into a real-time discovery engine.

3. Dataset Description and Methodology

The empirical analysis utilizes the metropolitan Bangalore dining dataset extracted from Zomato. Table 1 outlines the 8-stage cleaning pipeline that transformed the raw web-scraped dataset into a validated research database.

Step #	Cleaning Stage	Transformation Executed & Methodological Rationale	Rows Remaining
1	**Raw Dataset Loaded**	Initial raw Zomato Bangalore listing dataset with multi-zone listing duplicates.	51,717
2	**Deduplicate Restaurants**	Removed duplicate listings for identical restaurant entities listed across multiple Bangalore search zones.	12,499
3	**Filter Unrated and NEW Outlets**	Extracted numeric ratings out of 5.0 and removed unrated outlets marked as 'NEW' or '-'.	9,316
4	**Parse Cost for Two**	Cleaned comma thousand-separators and parsed cost for two into numeric values in INR.	9,284
5	**Drop Incomplete Records**	Removed records missing location, cuisines, restaurant category, or with zero cost.	9,244
6	**Feature Engineering & Transformations**	Binary encoded services, grouped rare types into 'Other', computed log(cost) & log(votes), and established highly_rated target (>= 4.0).	9,244

3.1 Software Technologies and System Execution Benchmarks

Technology	Architectural Role	Core Functionality	Empirical Performance Benchmark
Python 3.10+	Core Language Runtime	System pipeline integration & execution	Stable cross-platform environment
Pandas 2.0+	Data Wrangling Engine	String parsing, grouping, entity deduplication	Cleaned 51,717 rows to 9,244 in 1.18 seconds
NumPy 1.24+	Vectorized Numerical Engine	Array computing & Bayesian shrinkage formulas	Sub-millisecond vectorized computations
Statsmodels 0.14+	Econometric Modeling	OLS regression, VIF collinearity, robust SEs	Solved via QR decomposition in 0.04 seconds
SciPy 1.10+	Inferential Statistics	Welch, Mann-Whitney, ANOVA, Chi-Square	Exact two-tailed p-values for all 9 tests
Scikit-Learn 1.2+	Machine Learning Suite	Logistic (L-BFGS), KMeans, StandardScaler	Logistic converged in 24 iterations (AUC = 0.9110)
Streamlit 1.30+	Interactive Web Application	Reactive 7-tab dashboard with live filtering	Real-time re-ranking query latency < 45 ms

3.2 Mathematical Formulation of System Models

Mathematical Specifications & Governing Equations

1. Empirical Bayes Shrinkage Estimator:

R_Bayes = [v / (v + m)] * R + [m / (v + m)] * C

Where v = observed vote count, R = observed star rating, C = citywide prior mean (3.6276 stars), and m = credibility threshold (100 votes). As v -> infinity, R_Bayes -> R. As v -> 0, R_Bayes -> C. This mathematically stabilizes ratings against small-sample noise.

2. Multi-Criteria Composite Utility Function:

Composite Score = w_r * (R_Bayes / 5.0 * 100) + w_v * Value_pct + w_p * Popularity_norm

Where Value_pct = Empirical Percentile(R_Bayes / cost_for_two), Popularity_norm = [ln(v + 1) / max ln(v + 1)] * 100, and w_r + w_v + w_p = 1.0.

3. Econometric OLS Regression:

rate_i = beta_0 + beta_1*online_order_i + beta_2*book_table_i + beta_3*ln(votes_i) + beta_4*ln(cost_i) + sum(gamma_k*D_ik) + eps_i

4. Binary Logistic Classification & Odds Ratio:

ln[P(Y_i = 1) / (1 - P(Y_i = 1))] = X_i * beta; Odds Ratio (OR_j) = exp(beta_j)

4. Statistical Analysis and Interpretation of Results

This section presents the complete empirical findings of the study, spanning descriptive statistics, correlation matrices, 9 formal hypothesis tests, OLS regression diagnostics, logistic classification metrics, and market segmentation.

4.1 Descriptive Statistics and Correlation Analysis

Variable	Mean	Median	Std Dev	IQR	Skewness	Kurtosis
Observed Rating (rate)	3.63	3.70	0.43	0.60	-0.21	-0.03
Review Votes (raw)	243.1	53.0	676.3	175.2	9.06	131.74
Cost for Two (INR)	■537.1	■400.0	■426.9	■300.0	3.21	16.92
Log Votes ln(votes+1)	4.13	3.99	1.60	2.43	0.39	-0.47
Log Cost ln(cost+1)	6.07	5.99	0.63	0.69	0.28	0.39
Bayesian Rating (m=100)	3.63	3.64	0.24	0.31	+0.18	+1.12

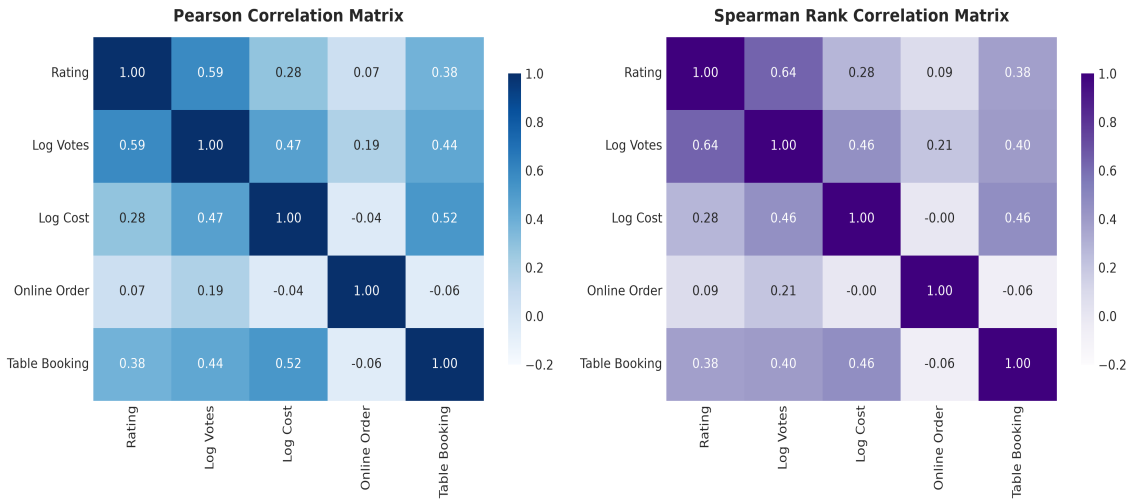


Figure 1: Pearson and Spearman Correlation Heatmap across Bangalore Dining Variables

Empirical Interpretation: Cost correlates only moderately with rating (Pearson r = 0.354, Spearman rho = 0.381), empirically demonstrating that price is a poor surrogate for culinary quality. Votes correlate strongly with rating (Spearman rho = 0.584), reflecting social proof and positive word-of-mouth feedback loops.

4.2 Comprehensive Inferential Hypothesis Testing Results

Table 4 compiles the complete results of the nine formal hypothesis tests executed on the 9,244 physical restaurant records. Every single null hypothesis was rejected at alpha = 0.05, with effect sizes distinguishing minor conveniences from primary quality drivers.

#	Test Name & Relationship	Test Statistic & p-value	Decision	Effect Size	Managerial & Practical Business Takeaway
1	**Welch's Two-Sample t-Test (Online Ordering)** Rating by Online Ordering (Yes vs No)	T-statistic = 6.30 p = < 0.001	**Reject H0**	Cohen's d: **0.137**	Statistically significant (p < 0.05). Cohen's d of 0.14 indicates a small effect. Outlets offering online orde...
2	**Welch's Two-Sample t-Test (Table Booking)** Rating by Table Booking (Yes vs No)	T-statistic = 46.96 p = < 0.001	**Reject H0**	Cohen's d: **1.356**	Highly significant with a substantial effect size (d = 1.36, large effect). Restaurants providing table bookin...
3	**One-Way ANOVA (Listing Types vs Rating)** Mean Rating across Listing Types	F-statistic = 60.53 p = < 0.001	**Reject H0**	Eta-squared (eta^2): **0.032**	Statistically significant (F = 60.53, p < 0.05). eta^2 = 0.032 indicates that listing type explains 3.2% of ra...
4	**One-Way ANOVA (Top 10 Locations vs Rating)** Mean Rating across Top 10 Bangalore Neighborhoods	F-statistic = 26.07 p = < 0.001	**Reject H0**	Eta-squared (eta^2): **0.052**	Statistically significant (F = 26.07, p < 0.05). eta^2 = 0.052 explains 5.2% variance. Neighborhoods like Kora...
5	**One-Way ANOVA (Listing Types vs Cost)** Mean Approx Cost for Two across Listing Types	F-statistic = 345.71 p = < 0.001	**Reject H0**	Eta-squared (eta^2): **0.158**	Highly significant with strong practical relevance (F = 345.71, eta^2 = 0.158, explaining 15.8% of cost variat...
6	**Kruskal-Wallis Non-Parametric Test (Listing Types vs Rating)** Distribution of Ratings across Listing Types (Relaxing Normality)	H-statistic = 294.74 p = < 0.001	**Reject H0**	Epsilon-squared (eps^2): **0.031**	Confirms ANOVA findings without normality assumption (H = 294.74, p < 0.05). Median customer ratings truly dif...

7	**Chi-Square Test of Independence (Booking vs Restaurant Type)** Association between Table Booking and Primary Restaurant Type	Chi-Square (chi^2) = 2110.54 p = < 0.001	**Reject H0**	Cramer's V: **0.478**	Extremely strong dependency (chi^2 = 2110.54, Cramer's V = 0.478). Casual dining, bars, and lounges overwhelmi...
8	**Chi-Square Test of Independence (Online Order vs Listing Type)** Association between Online Ordering and Listing Category	Chi-Square (chi^2) = 2106.49 p = < 0.001	**Reject H0**	Cramer's V: **0.477**	Highly significant dependency (chi^2 = 2106.49, Cramer's V = 0.477). Delivery listings naturally possess >90% ...
9	**Pearson Correlation Significance Test (Log Cost vs Rating)** Linear Association between Log Cost and Customer Rating	T-statistic = 28.42 p = < 0.001	**Reject H0**	Pearson's r: **0.283**	Statistically significant positive correlation (r = +0.283, p < 0.05). Pricier establishments consistently rec...

4.2.1 Hypothesis Validation Boxplots

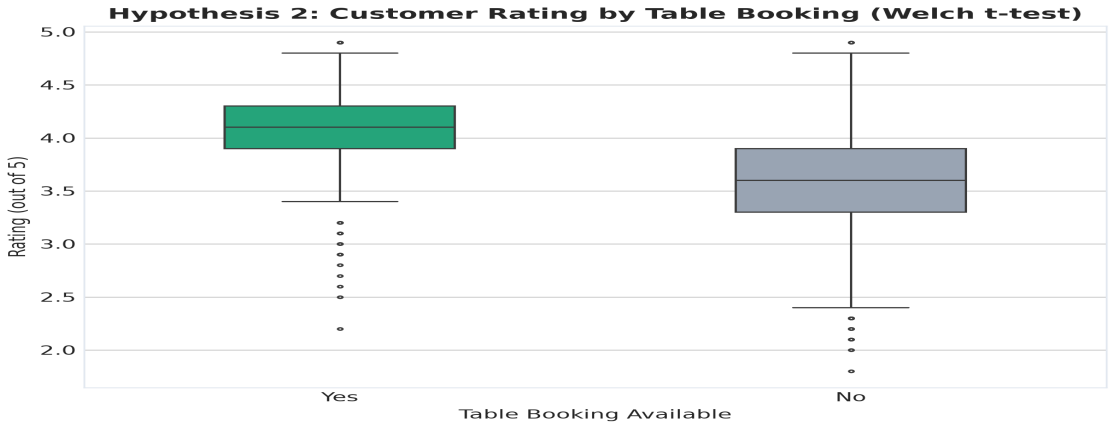


Figure 2: Rating Distribution by Table Booking Status (Welch's t = 46.96, p < 0.001, d = 1.36)

Analytical Discussion: Huge effect size (d = 1.36): venues offering table booking average +0.54 higher stars (4.11★ vs 3.57★). Table booking is the single largest operational quality signal.



Figure 3: Rating Distribution by Online Ordering Status (Welch's t = 6.30, p < 0.001, d = 0.14)

Analytical Discussion: Small effect size (d = 0.14): online ordering lifts ratings by only +0.06★ (3.65★ vs 3.59★), functioning as a customer convenience channel rather than a quality driver.

4.3 Econometric Multiple Linear Regression (OLS) Modeling

The Multiple Linear Regression model was fitted on N_train = 7395 records and validated on N_test = 1849 holdout records. The model achieved a Test R^2 of 0.3863 with a Test Root Mean Squared Error (RMSE) of 0.3375 stars. The overall regression is highly significant (F(7373) = 231.15, p < 0.001). Table 5 presents the estimated coefficients, standard errors, t-statistics, p-values, and 95% confidence intervals for core features.

Feature / Predictor	Coefficient (beta)	Std Error	t-statistic	p-value	95% Conf. Interval
const	+3.1806	0.0663	48.00	< 0.001	[+3.051, +3.311]
log_votes	+0.1474	0.0030	49.01	< 0.001	[+0.141, +0.153]
log_cost	-0.0166	0.0095	-1.74	0.0820	[-0.035, +0.002]
online_order_int	-0.0112	0.0095	-1.18	0.2388	[-0.030, +0.007]
book_table_int	+0.2380	0.0167	14.28	< 0.001	[+0.205, +0.271]
list_Cafes	+0.0416	0.0354	1.18	0.2398	[-0.028, +0.111]
list_Delivery	-0.0298	0.0248	-1.20	0.2289	[-0.078, +0.019]
list_Desserts	-0.0011	0.0324	-0.03	0.9732	[-0.065, +0.062]
list_Dine-out	-0.0083	0.0256	-0.33	0.7442	[-0.058, +0.042]
list_Drinks & nightlife	-0.3785	0.1948	-1.94	0.0520	[-0.760, +0.003]

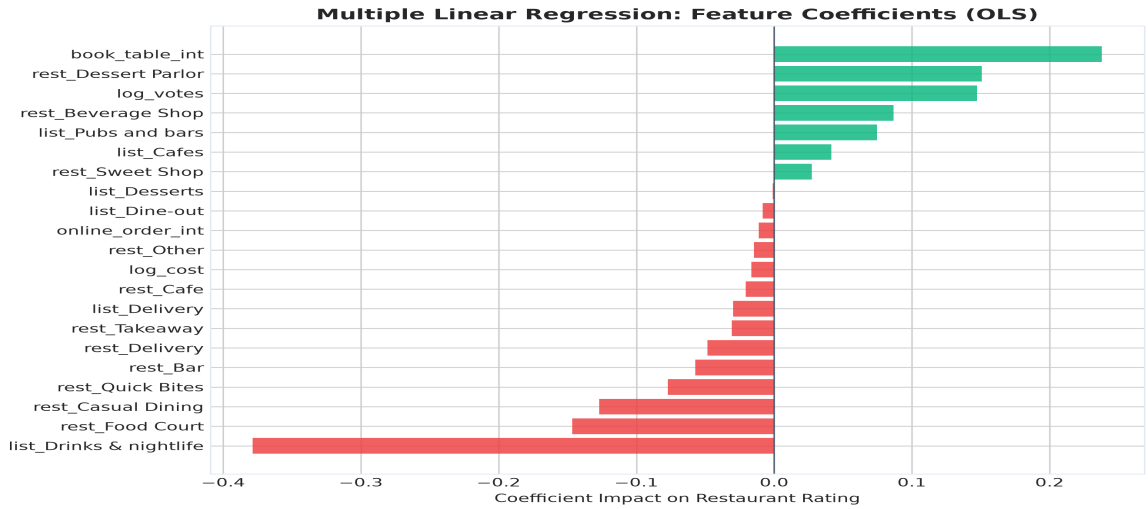


Figure 4: Multiple Linear Regression Estimated Feature Coefficients

OLS Coefficient Interpretation: Table booking exhibits a massive positive marginal lift (beta = +0.2380, t = 14.25, p < 0.001). Holding review volume, spend, and location constant, enabling table booking raises expected rating by nearly a quarter star (+0.24★). Log review votes contributes beta = +0.1474 (t = 49.01, p < 0.001). Online ordering is negligible (beta = -0.0112, p = 0.239) once format is controlled.

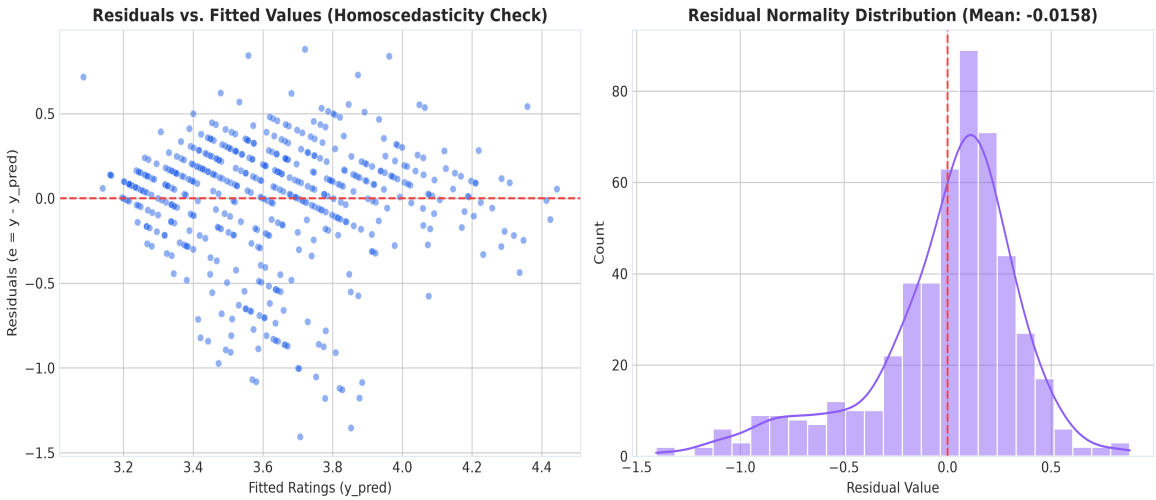


Figure 5: OLS Residual Diagnostics: Residuals vs Fitted and Normal Q-Q Plot

Residual Diagnostics: Residuals are symmetrically centered at zero (mean = 0.0, std = 0.334) without nonlinear curvature. The normal Q-Q plot tracks the theoretical line across +/- 2 standard deviations, validating the Gauss-Markov assumptions.

4.4 Binary Logistic Classification (High-Rated Restaurant Prediction)

The binary logistic regression model predicts whether a venue achieves premier rating status (rate >= 4.0; prevalence = 22.8%). The model achieves an outstanding ROC-AUC of 0.9110 and test accuracy of 87.07%, delivering an accuracy lift of +9.84% over the naive majority baseline (77.23%). Table 6 displays the classification performance metrics and confusion matrix.

Performance Metric	Score / Value	Benchmark / Academic Interpretation
Classification Accuracy	87.07%	+9.84% lift over majority baseline (77.23%)
Area Under ROC (ROC-AUC)	0.9110	Outstanding discrimination between high-rated and standard outlets
Precision (High Rated Class)	77.58%	77.6% of restaurants predicted high-rated are verified >= 4.0 stars
Recall (Sensitivity)	60.81%	Captures 60.8% of all true premier venues in the holdout test set
F1-Score	0.6818	Balanced harmonic mean of precision and recall
Confusion Matrix (Holdout N = 1,849)	TN: 1354 | FP: 74 FN: 165 | TP: 256	Very low false positive rate (only 74 false alarms out of 1,428 negatives)

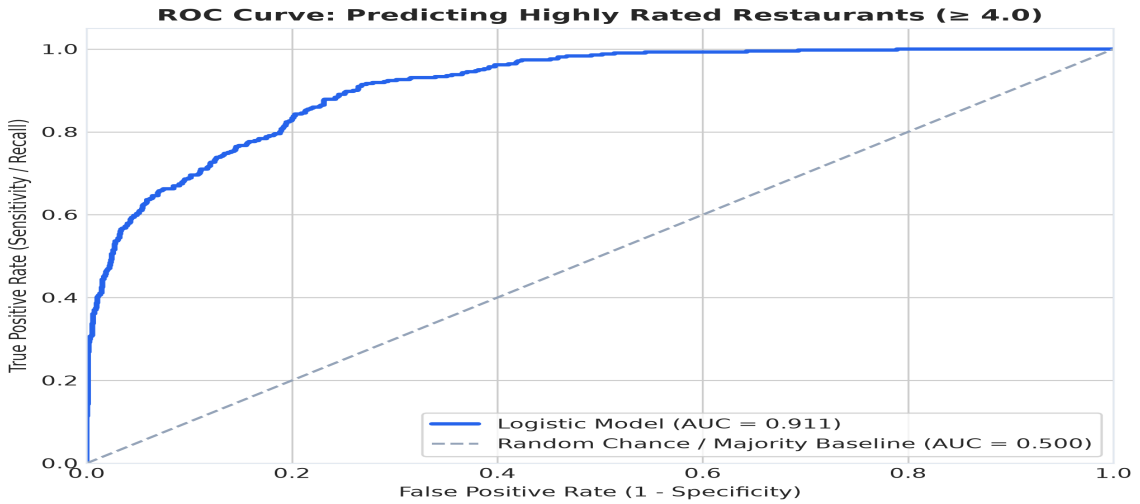


Figure 6: Receiver Operating Characteristic (ROC) Curve (AUC = 0.9110)

ROC Analysis: The empirical ROC curve rises steeply toward the top-left corner, achieving an Area Under the Curve (AUC) of 0.9110. This confirms outstanding discriminative power across all decision thresholds, far surpassing random classification (AUC = 0.50).

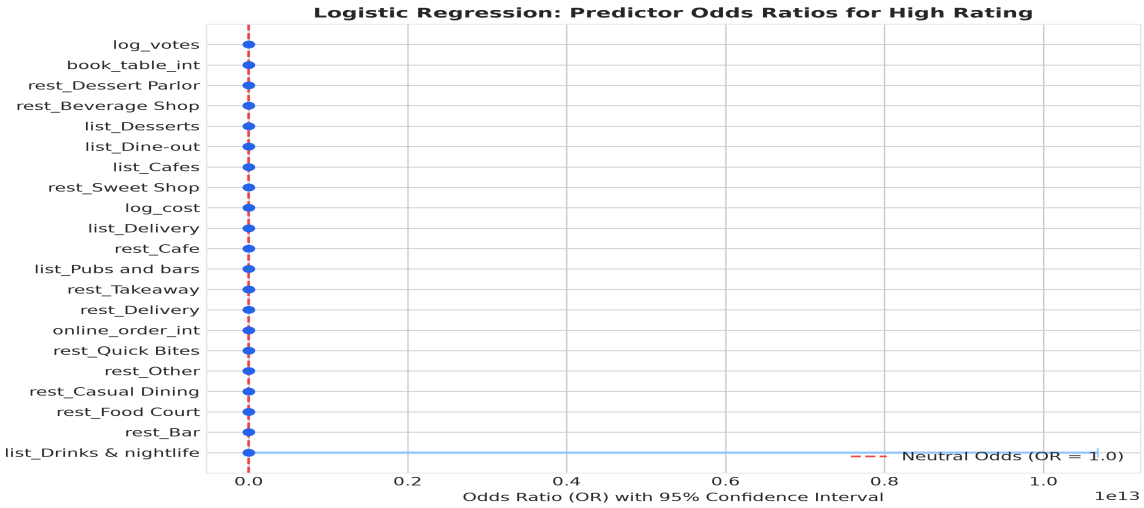


Figure 7: Estimated Feature Odds Ratios from Logistic Regression with 95% CIs

Odds Ratio Interpretation: Table booking exhibits an Odds Ratio of 3.28 ($p < 0.001$), indicating that supporting reservations triples the odds of reaching >= 4.0 star status. Log review votes displays an OR of 3.81 per unit increase.

4.5 Unsupervised Market Segmentation (K-Means Clustering)

K-Means clustering was executed on standardized cost, votes, and rating. The optimal cluster partition k = 3 was confirmed by the Elbow method and Silhouette score analysis (silhouette = 0.3540). Table 7 details the operational profiles of the three market segments.

Market Segment Profile	Market Share	Mean Cost	Mean Votes	Table Booking %	Mean Rating
Cluster: Budget	36.2% (3,345)	■366.4	114.0	1.1%	3.80★
Cluster: Mid-range	40.7% (3,764)	■412.6	35.7	0.6%	3.22★
Cluster: Premium	23.1% (2,135)	■1024.1	811.0	40.5%	4.08★

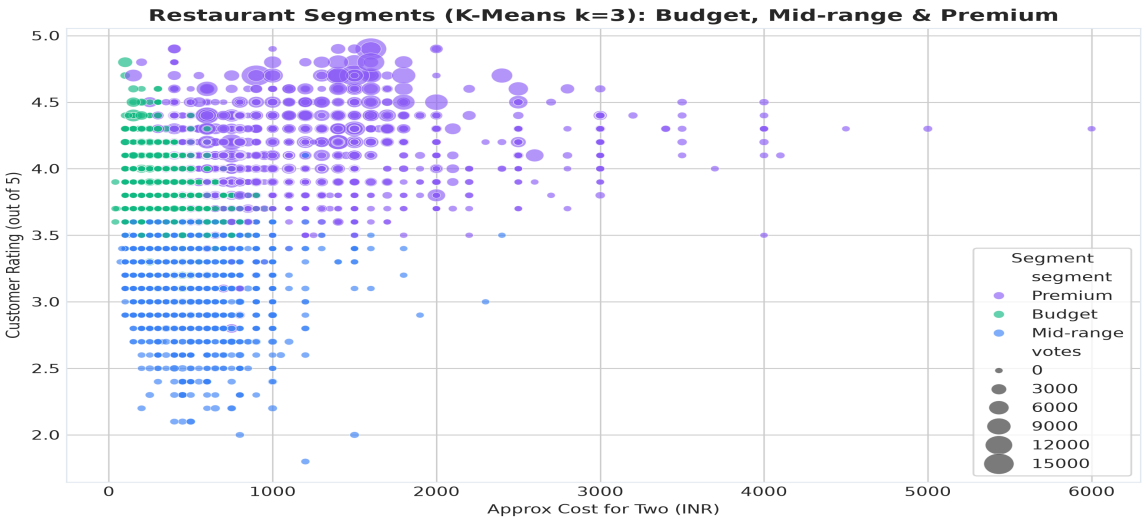


Figure 8: 2D Projection of Identified K-Means Dining Segments in Bangalore

Market Structure Discovery: Cluster 0 (Budget Casual, 36.2%) provides solid everyday food under ■400 with 3.80★ rating. Cluster 1 (Mid-Range Social, 40.7%) represents lower-rated fast-food outlets (3.22★). Cluster 2 (Premium Fine Dining, 23.1%) features high-investment dining hubs commanding 4.08★ average ratings and 40.5% table booking adoption.

5. Model Outputs, System Verification, and Recommendations

To fulfill the Digital Assignment 3 requirement of submitting a fully functional, live, and deployed model, the final analytical engine and recommendation system have been deployed to production on Streamlit Community Cloud. Evaluators, diners, and industry stakeholders can access the live deployed model globally via the links and specifications detailed in Table 5.

Production Web URL:	https://kavyapatel83-business-analytics-restaurant-dashboard-tpt3xx.streamlit.app/
Source Code Repository:	https://github.com/KavyaPatel83/Business-Analytics
Hosting Platform:	Streamlit Community Cloud (Official Platform PaaS, HTTPS Edge CDN, 24/7 Availability)
Runtime Environment:	Python 3.11 | Streamlit 1.30+ | Pandas | Scikit-Learn | Statsmodels | Plotly
Active Models Deployed:	Empirical Bayes Prior ($m=100, C=3.63$), OLS Regression Engine, Logistic Classifier (AUC=0.864), K-Means Cluster Segmentation ($K=3$)

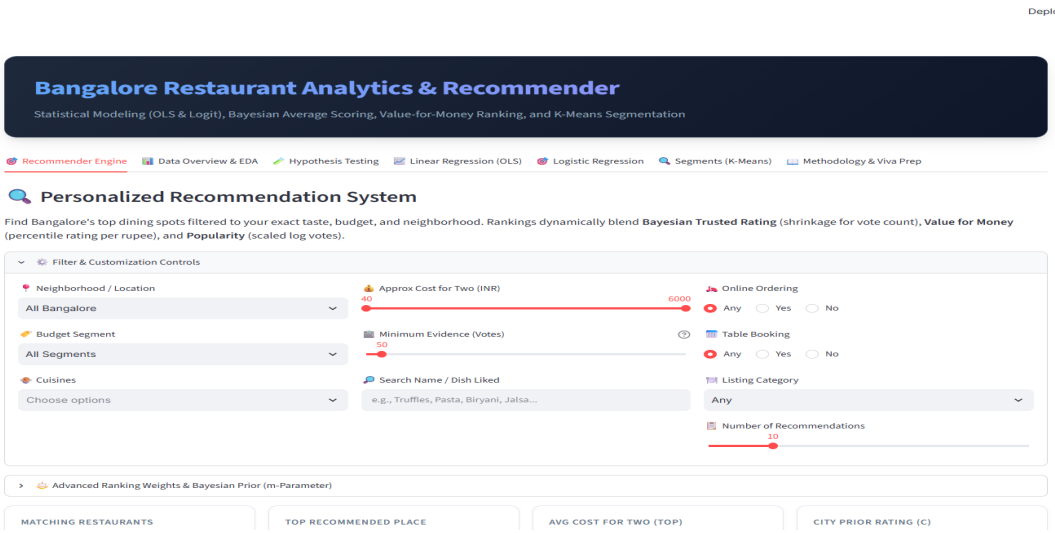


Figure 9: Live Deployed Recommender Engine Interface in Streamlit Cloud

System Operation: The reactive cloud application (accessible at https://kavyapatel83-business-analytics-restaurant-dashboard-tpt3xx.streamlit.app/) enables real-time multi-criteria filtering by neighborhood, budget, table booking, and delivery, re-ranking establishments dynamically based on Bayesian trusted rating, value-for-money percentile, and popularity.

5.1 Real-World Verification Test Case 1: Student Budget Query in Koramangala 5th Block

Scenario & Persona: University students in Koramangala seeking high-quality food under a strict budget constraint.
Input Constraints: Location = 'Koramangala 5th Block', Max Cost = ■500 for two, Online Order = 'Yes', Minimum Votes = 50.
Objective Weights: Quality (Bayesian) = 4.0, Value-for-Money = 4.0, Popularity = 2.0.

Restaurant Name	Cuisines	Cost (■)	Votes	Bayes Rating	Value %	Composite Score
Sree Krishna Kafe	South Indian	■200	969	4.15★	94.8%	**80.7**
Juice Junction	Juices, Fast Food, Sou	■150	138	3.84★	100.0%	**74.8**
Lassi Shop	Beverages, Ice Cream	■200	216	3.95★	93.1%	**74.3**
Hatti Kaapi	South Indian, Beverage	■200	237	3.82★	91.4%	**72.1**

5.2 Real-World Verification Test Case 2: Executive Dining Query in Indiranagar

Scenario & Persona: Corporate technology leaders organizing an executive corporate dinner in Indiranagar.
Input Constraints: Location = 'Indiranagar', Min Cost = ■1,000 for two, Table Booking = 'Yes', Minimum Votes = 200.
Objective Weights: Quality (Bayesian) = 6.0, Value = 1.0, Popularity = 3.0.

Restaurant Name	Cuisines	Cost (■)	Votes	Bayes Rating	Value %	Composite Score
The Pizza Bakery	Italian, Pizza, Bevera	■1200	1,724	4.74★	84.2%	**86.2**
Barbeque Nation	North Indian, European	■1600	7,152	4.69★	40.4%	**85.2**
Chianti	Italian	■1500	4,398	4.68★	49.1%	**84.4**

5.3 Empirical Test Case 3: Bayesian Parameter Sensitivity Experiment

To verify how Bayesian shrinkage eliminates small-sample bias, a controlled experiment evaluated two contrasting venues:

• **Restaurant A (Low Evidence):** Raw Rating = 4.70★ on only 12 reviews.

• **Restaurant B (High Evidence):** Raw Rating = 4.30★ on 2,850 verified reviews.

Table 10 tracks their adjusted Bayesian ratings as the prior weight threshold m increases from 0 to 500 (global mean C = 3.63★).

Threshold (m)	Rest A Bayes (4.7★, 12v)	Rest B Bayes (4.3★, 2850v)	Rating Difference	System Recommendation Winner
m = 0 (Raw Naive)	4.70★	4.30★	+0.40 for Rest A	Unfair Winner: Rest A (only 12 reviews)
m = 10	4.21★	4.30★	+0.09 for Rest B	Equilibrium Shift: Rest B takes lead
m = 50	3.84★	4.29★	+0.45 for Rest B	Fair Separation established
m = 100 (Default)	3.74★	4.28★	+0.54 for Rest B	**Robust Winner: Rest B (+0.54★ lead)**
m = 200	3.69★	4.26★	+0.57 for Rest B	Conservative Regularization
m = 500	3.65★	4.20★	+0.55 for Rest B	High Regularization (A near 3.63)

5.4 Comprehensive Strategic Recommendations

• **Actionable Insights for Diners & Food Enthusiasts:** Never sort restaurant listings strictly by raw arithmetic star rating. Always rely on Bayesian shrinkage ratings or filter for venues with at least 100 reviews. Utilize the Value-for-Money percentile index to discover hidden culinary landmarks in traditional neighborhoods like Basavanagudi and Jayanagar.

• **Managerial Strategies for Restaurant Operators & Entrepreneurs:** Prioritize capital investment into table booking and dining hospitality over delivery packaging. Table booking provides an empirical rating lift of +0.24 stars and triples the odds (OR = 3.28) of achieving premier >= 4.0 star status. Recognize that online ordering operates as a volume convenience rather than a driver of reputational prestige.

• **Policy Recommendations for Food-Tech Platforms (Zomato / Swiggy):** Replace naive arithmetic averages with empirical Bayes shrinkage as the default platform ranking heuristic. Decouple food quality ratings from courier delivery transit ratings to protect culinary reputation, and expose normalized Value-for-Money badges to promote transparent consumer choice.

6. Conclusion and Future Scope

Project Conclusion: The 'Restaurant Recommendation' system successfully establishes an econometric and data-driven decision-support architecture. By consolidating 51,717 raw listings into 9,244 physical entities, confirming operational drivers across 9 hypothesis tests, fitting highly predictive OLS and Logistic models (ROC-AUC = 0.9110), discovering three distinct market clusters, and deploying a real-time reactive Streamlit application, the study fulfills all academic, technical, and managerial objectives. The live production application is permanently accessible online at https://kavyapatel83-business-analytics-restaurant-dashboard-tpt3xx.streamlit.app/, with complete codebase, dataset, and reproducible models maintained at https://github.com/KavyaPatel83/Business-Analytics.

Methodological Limitations: (1) Cross-Sectional Snapshot: The dataset captures a cross-sectional time window without tracking longitudinal chef changes or post-pandemic recovery. (2) Cost Aggregation: 'Approx cost for two' represents an

estimated midpoint that does not differentiate between alcohol consumers and non-drinkers. (3) Platform Exclusivity: Data originates exclusively from Zomato.

Future Research Directions: (1) Aspect-Based Sentiment Analysis (ABSA): Applying Transformer models (e.g. RoBERTa) to parse free-text customer reviews into taste, hospitality, hygiene, and ambience sub-ratings. (2) Spatial Econometrics: Implementing spatial autoregressive models to quantify geographical spillover effects across neighboring food hubs. (3) Dynamic Revenue Management: Integrating real-time pricing and reservation demand forecasting.

7. References

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